

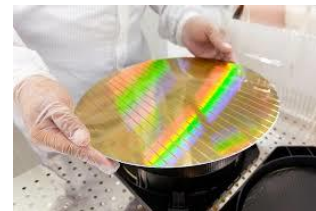
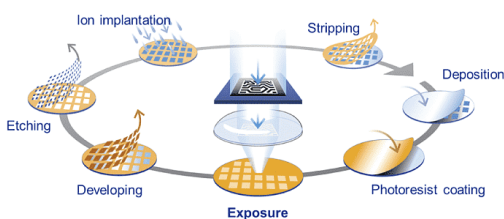
Advancing Semiconductor Fabrication: A CNN-Based Wafer Defect Detection with XAI Insights

Paper Id: 821

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Chip manufacturing is **complex**

Several opportunities for a defect to occur



> 1000
Processing steps to
finish a chip

> 10
Weeks of
processing time

Start

End

> 100

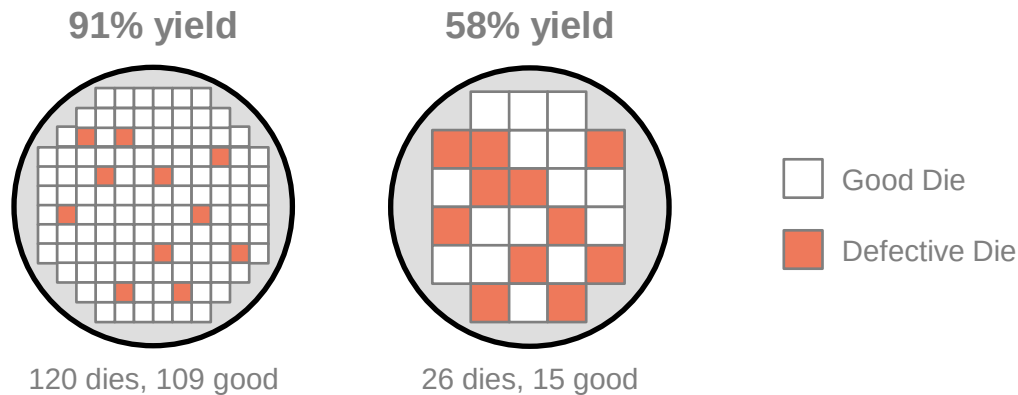
Equipment used in
manufacturing



> 40000

wafers started every month

Yield maximization is critical for **cost reduction**



↓ Defects = ↑ Yield = \$ Cost ↓



Wafer defect analysis is mainly performed manually by expert engineers, attaining accuracy rates below 45% [1].

Challenges

- Time-consuming
- Varying expertise
- Strict IP policies

Problem Statement

Automated Semiconductor Wafer Defect Classifier

Use a neural network to **identify** wafers with single defect patterns and **classify** them into different defect groups with >90% accuracy

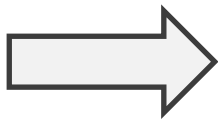


Literature review

Approach	Details	Limitations
Manual Inspection	Inefficient, time-intensive, prone to human errors and accuracy rates below 45% [1].	• Ineffective for high-volume manufacturing processes. • Prone to subjective errors due to manual assessment. • Limited scalability
Machine Learning	Cheon et al. (2019): CNN with k-NN for unknown defects [2]. Zheng et al. (2021): DCNN for wafer patterns. - Outperformed traditional methods, Achieved 93.75% accuracy [3].	Cheon: Poor generalization in noisy environments. Zheng: Difficulty with detecting localized failure patterns ("Loc").
Deep Learning	Saqlain et al. (2020): Deep-layered CNN for imbalanced datasets. - Focused on WM-811K dataset [4]. Kyeong & Kim (2018): CNN for mixed-type defects. - Addressed mixed-class issues [5].	Saqlain: Significant performance gaps for rare defect categories. Kyeong & Kim: Limited accuracy for overlapping defect categories.

Limitations in Existing Research

Data Challenges



Severe class imbalance: “None” class dominates most datasets

Inconsistent image dimensions and quality:
Complicates preprocessing steps

Model Performance



Difficulty achieving high accuracy for **multiple wafer defect** classes

Overfitting risks: Insufficient regularization leading to performance drops

Explainability



Lack of comprehensive methods to **interpret** and **justify** model decisions

Notable Contribution of This Work

ResNet50 achieved superior performance (~99% accuracy) across multiple wafer defect classes in the WM-811k dataset, including Donut, Edge-Ring, Near-full, and Random

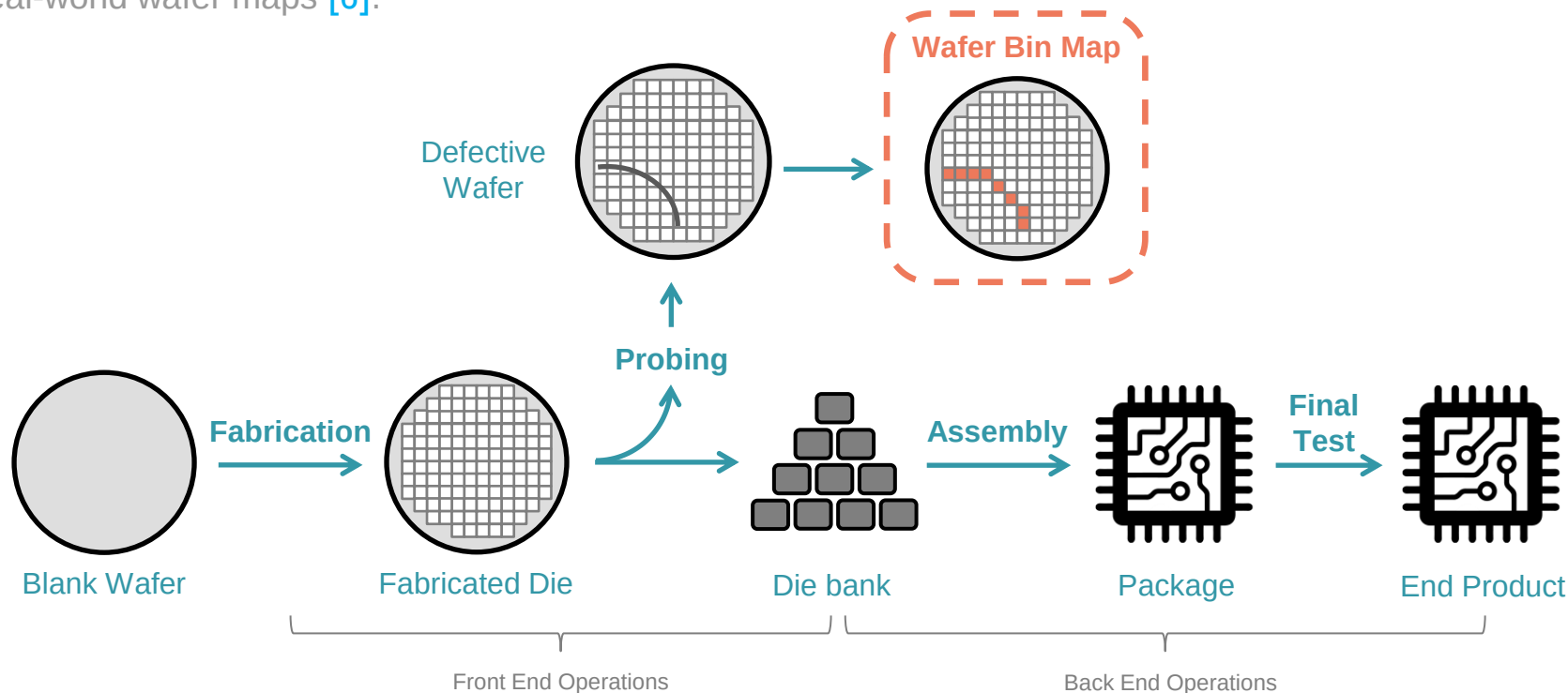
Addressed class imbalance through advanced data augmentation

In addition to ResNet-50, a custom CNN, Inception V3, and SVM classifiers were developed and evaluated for wafer defect detection

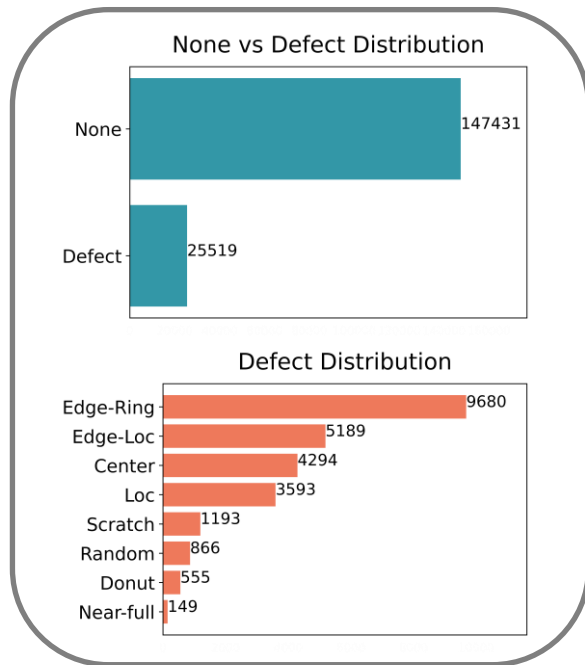
XAI-Grad-CAM visualizations validated model focus on critical defect areas and improved transparency in model decision-making processes.

Where does our **data** come from?

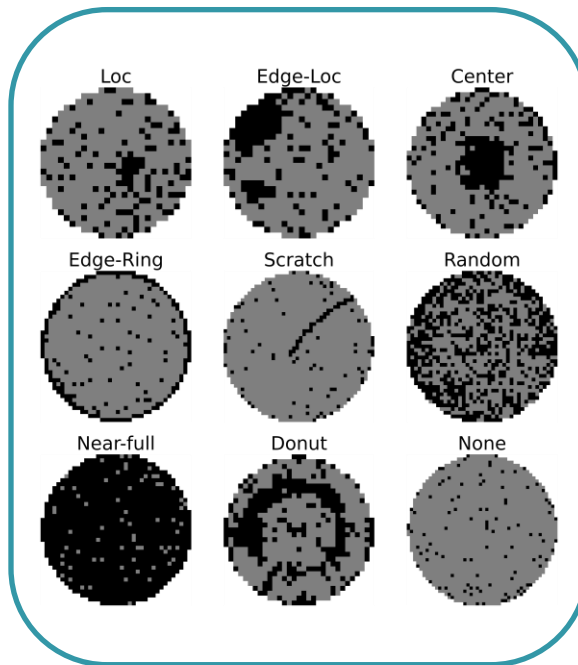
WM-811K is one of the largest open source dataset from real production process, comprising 811,457 real-world wafer maps [6].



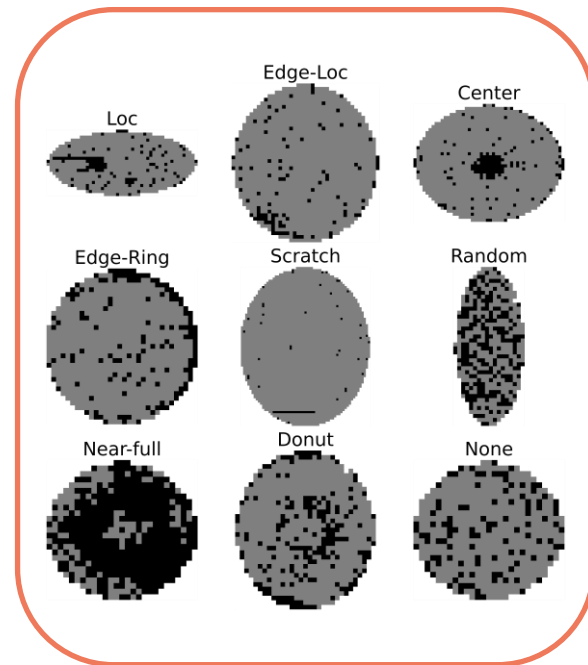
What does our data **look** like?



Highly Imbalanced
Undersampling, data augmentation

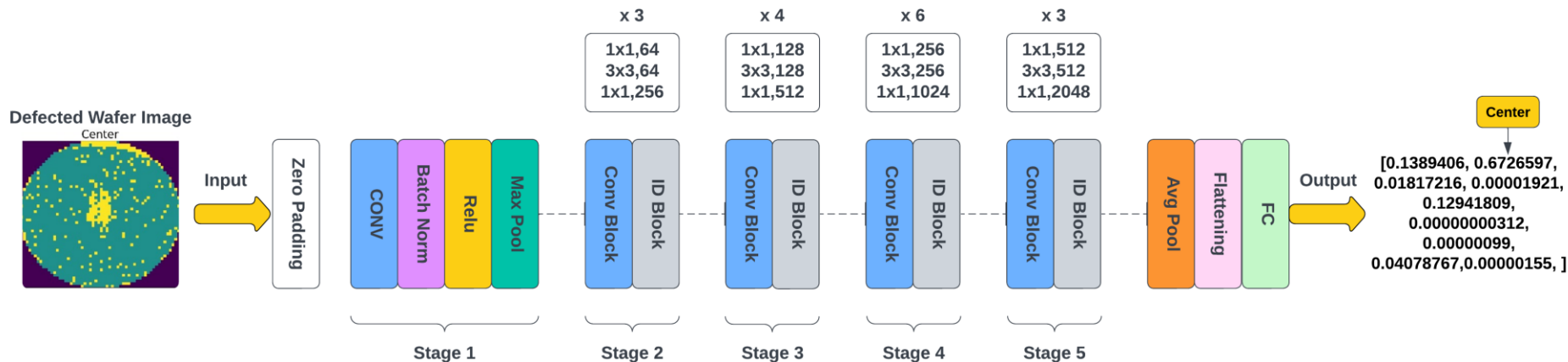


Single-defect multi-class
9 classes total, including none



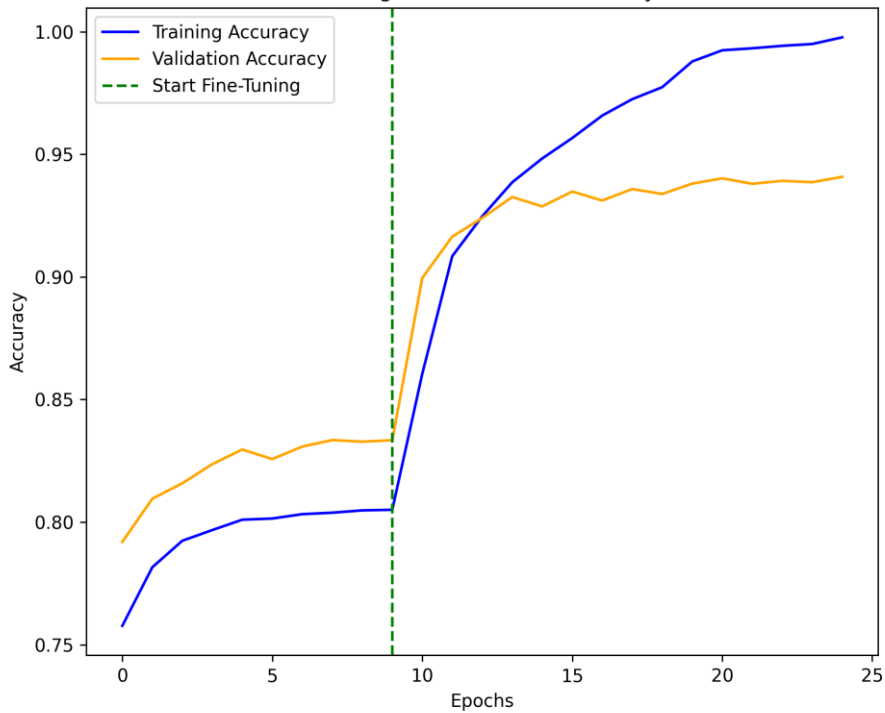
Variable quality and resolution
Resizing and pre-processing

Proposed ResNet-50 CNN model architecture

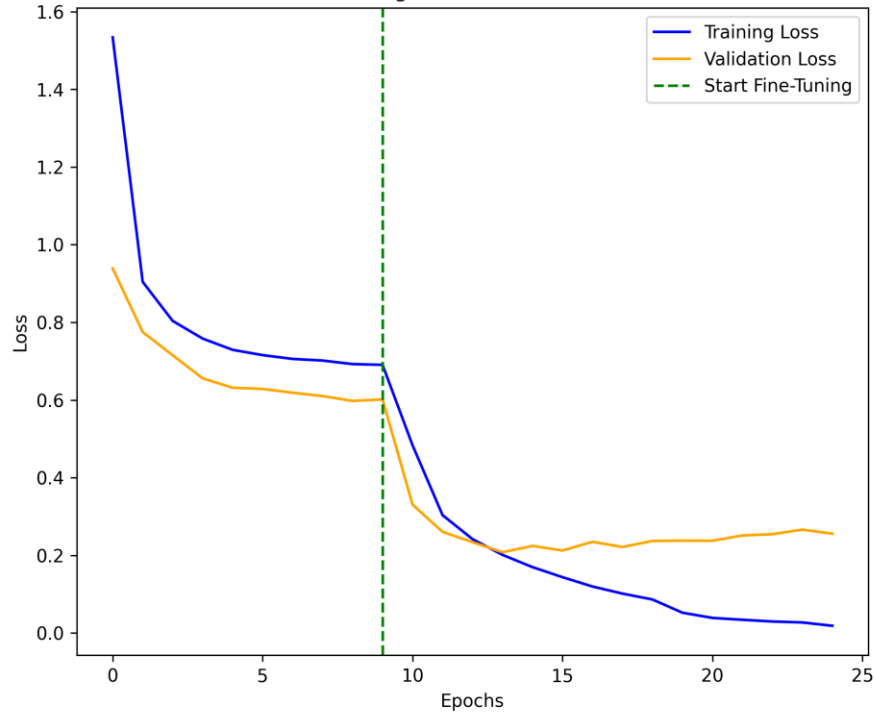


Model architecture exploration

Training and Validation Accuracy

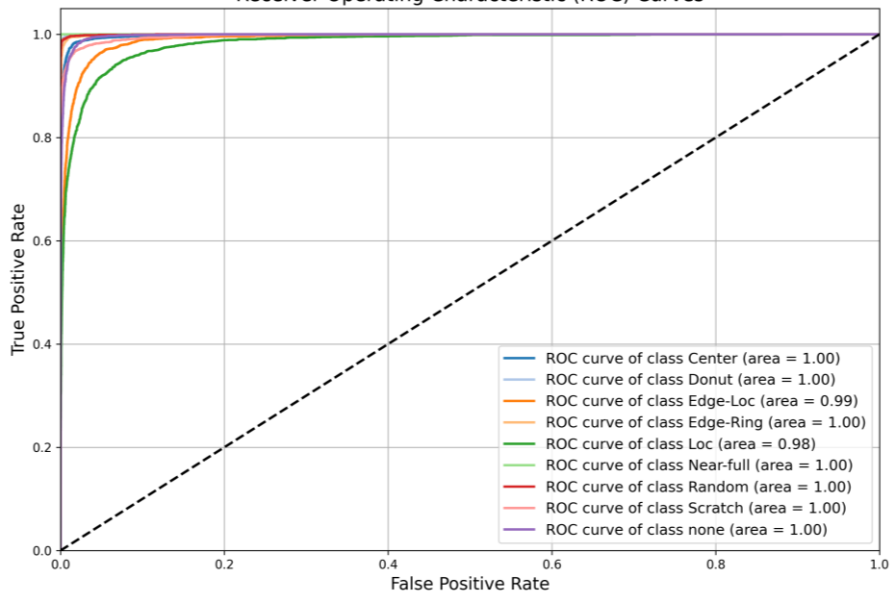


Training and Validation Loss



Model architecture exploration

Receiver Operating Characteristic (ROC) Curves



Confusion Matrix



Per-class Performance Metrics of the ResNet-50 CNN model

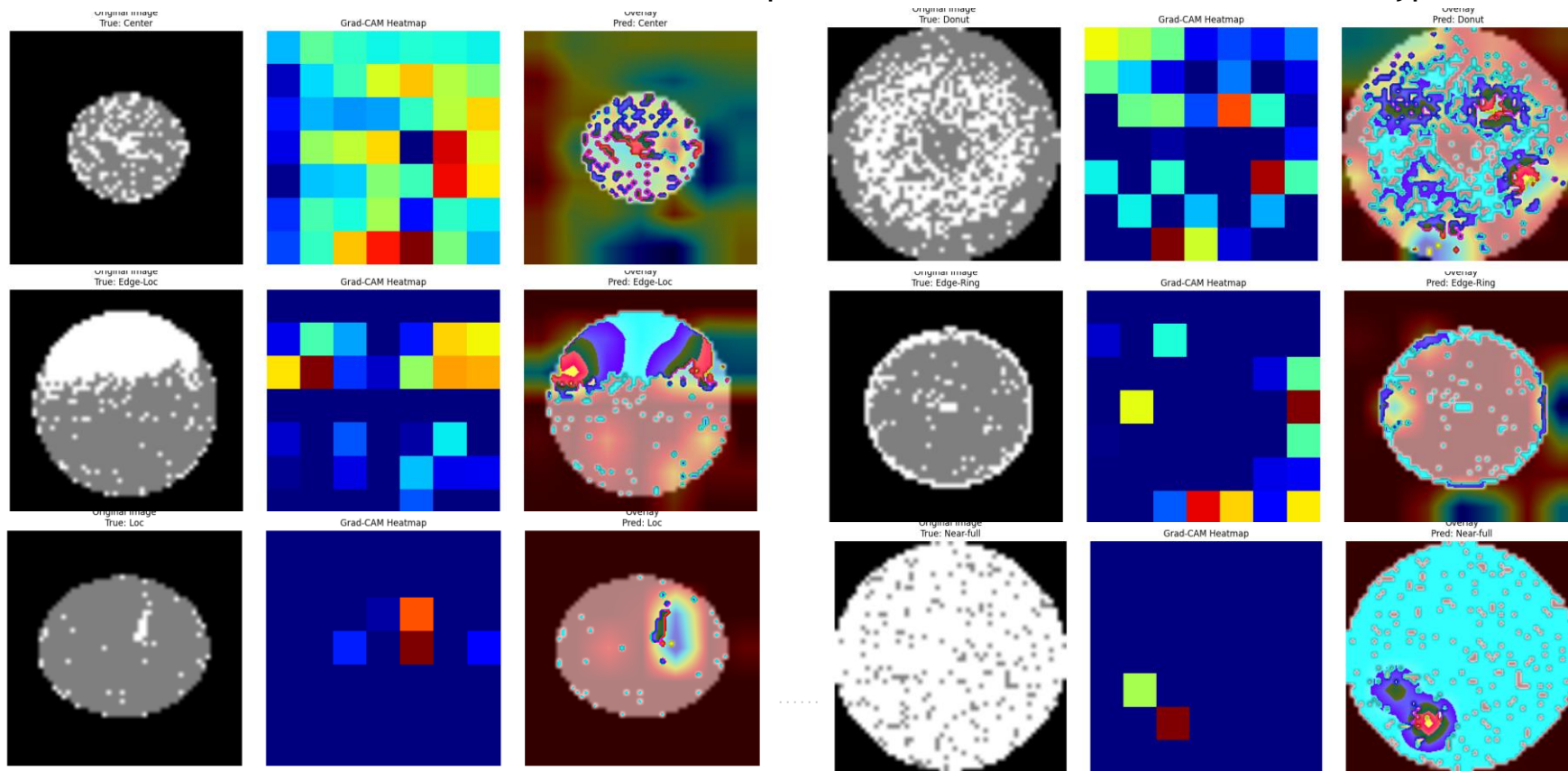
Class	Accuracy	Precision	Recall	F1-score
Center	0.9859	0.9346	0.9414	0.9380
Donut	0.9953	0.9758	0.9830	0.9794
Edge-Loc	0.9677	0.8562	0.8577	0.8570
Edge-Ring	0.9938	0.9671	0.9787	0.9728
Loc	0.9577	0.8472	0.7617	0.8022
Near-full	0.9993	0.9963	0.9977	0.9970
Random	0.9964	0.9837	0.9840	0.9838
Scratch	0.9846	0.9268	0.9373	0.9321
none	0.9843	0.8928	0.9542	0.9225

Comparative Analysis of Our Other Proposed Models

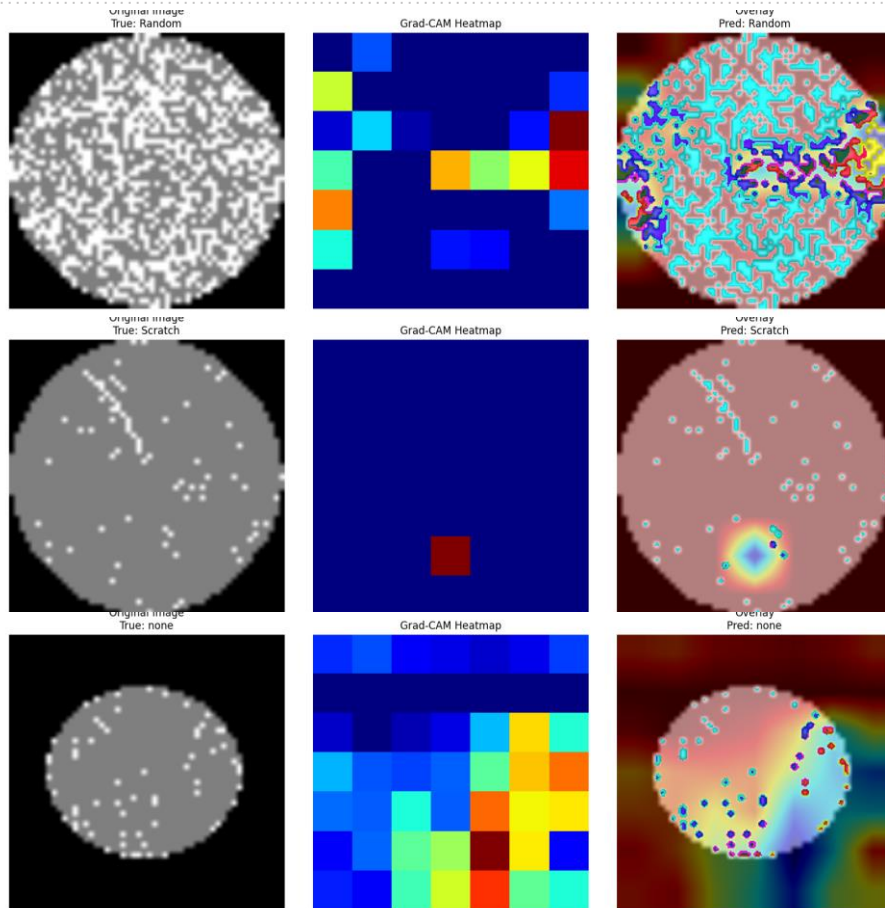
Classifier	Accuracy	Precision	Recall	F1 Score
ResNet50	0.9408	0.9403	0.9409	0.9405
Custom-CNN	0.8621	0.8758	0.8641	0.8624
Inception v3	0.8816	0.8842	0.8822	0.8783
SVM	0.8230	0.8182	0.8076	0.8118

What does our model **see**?

XAI-Grad-CAM: Visualizes the class activation maps for the different defect classification types



What does our model **see**?



Production model **pipeline**

Customized data pipeline to meet the business need with real time inference



Consistent



Time Efficient



Customizable



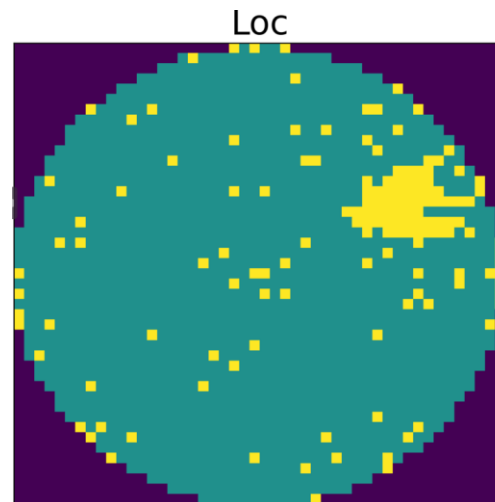
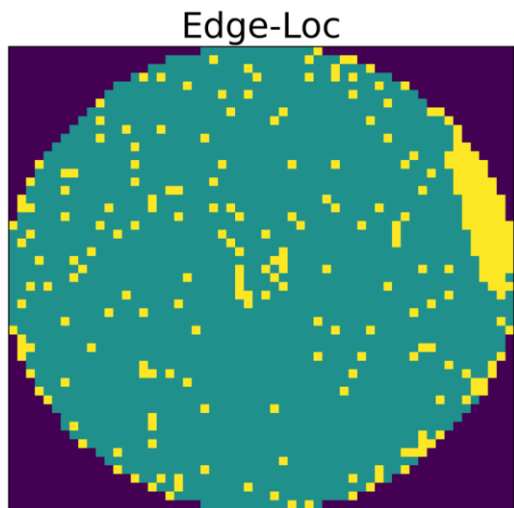
Cost Effective



Wafer Map Classifier for Early
and Fast **Defect Detection** and
Defect Source Identification

Model can be tuned for
custom **Defect Patterns** for
fast defect sourcing

Future Research Direction



Future research could explore alternative model architectures, implement targeted training strategies, or employ advanced data augmentation techniques to better the detection of the **Edge-Loc** and **Loc** defect class on wafers.

References

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- [1] A. Drozda-Freeman, M. McIntyre, M. Retersdorf, C. Wooten, X. Song, and A. Hesse, "The application and use of an automated spatial pattern recognition (spr) system in the identification and solving of yield issues in semiconductor manufacturing," in Proceedings of the 2007 IEEE/SEMI Advanced Semiconductor Manufacturing Conference, Stresa, Italy, Jun. 2007, pp. 302-305. [Online]. Available: <https://doi.org/10.1109/ASMC.2007.4391932>
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- [6] M. -J. Wu, J. -S. R. Jang and J. -L. Chen, "Wafer Map Failure Pattern Recognition and Similarity Ranking for Large-Scale Data Sets," in IEEE Transactions on Semiconductor Manufacturing, vol. 28, no. 1, pp. 1-12, Feb. 2015, doi: 10.1109/TSM.2014.2364237



Thanks for Watching
Any Questions?